

Supplementary Material for “Motion Planner for Unmanned Ground Vehicles based on a Lexicographic Genetic Algorithm”

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Abstract

This is supplementary material that gives more details on the proposed motion planner for UGVs using a lexicographic genetic algorithm, both regarding the methodology and the experiments. The first one is related to the path-decoding procedure that transforms candidate solutions represented in a polar, motion-based format into Cartesian task-space trajectories. Each of the path segments is represented in this form by a relative turning angle and a segment length, so that the decoded trajectory would retain the motion behavior of a differential-drive vehicle. The document also describes how to calculate cumulative headings, terminal cut points, and corrected final segments to produce feasible paths to determine the objective function and constraints. Second, additional simulation results are provided for three planning tasks: visual comparisons of optimal trajectories, boxplots of terminal docking error, path length, undulation, and number of segments. Lastly, the Kruskal–Wallis test and Dunn–Holm post hoc comparisons are reported to assess the significance of performance differences between algorithm configurations. These supplementary results confirm the consistency and repeatability of the proposed planner and provide further evidence of its ability to generate motion-consistent UGV paths.

1 Path-Decoding Procedures of the Motion-Planning Problem

The solutions Ψ generated by the method describe UGV paths in polar, motion-based representation by using relative turning angles and segment lengths. However, the geometrical properties of the objective functions and constraints require calculations in task space (i.e., with Cartesian coordinates). This operation is called path encoding.

A candidate solution is composed of n path segments. Each segment s is described by a relative turning angle θ_i and a segment length l_i . The first segment of the path corresponds to the home-base h , whose pose is defined by its Cartesian coordinates and initial heading. Starting from the home-base pose, each subsequent segment is obtained by updating the current heading based on the relative turning angle, then translating the vehicle by the corresponding segment length [1].

1.1 From Motion Representation to Task Space

The polar representation ζ^p of a solution contains the sequence of relative turning angles and segment lengths. The decoded representation ζ^c contains the Cartesian coordinates of the path nodes. Since a path with n segments contains the home node and one endpoint for each segment, ζ^p is represented with $(n+1)$ rows and two columns.

Algorithm S.1 shows the decoding procedure. It takes the path representation in motion space ζ^p , the maximum number of segments n , and the home-base pose h as input. It returns the path in task space ζ^c together with the cumulative heading vector Φ . The cumulative heading vector is used for procedures, such as terminal-orientation and turning-direction checking.

The procedure follows the motion behavior of a differential-drive UGV at the planning level: the vehicle first changes its heading relative to the previous segment, then moves forward along a straight segment. Therefore, each decoded node is not an independent waypoint, but the result of a sequence of motion-consistent commands. This is important because the optimizer does not search directly over arbitrary Cartesian waypoints. Instead, it searches over relative heading changes and travel distances, which gives the decoded path an explicit motion interpretation.

For each segment s , the cumulative heading is updated recursively, with `wrapToPi` constraining the angle to the interval $[-\pi, \pi]$. The equation is defined as in Eqn. (1).

$$\phi_i = \text{wrapToPi}(\phi_{i-1} + \theta_i) \quad (1)$$

Algorithm S.1: Path Transformation from Motion Space to Task Space

input: ζ^P , a matrix $(n \times 2)$ of a path in motion space
input: n , maximum number of path segments
input: h , a vector (1×3) with the home-base pose (x_h, y_h, ϕ_h)
output: ζ^c , a matrix $((n+1) \times 2)$ of the path in task space
output: Φ , a vector $(1 \times (n+1))$ of cumulative headings

```
1 begin
2    $\zeta^c \leftarrow 0_{(n+1) \times 2}$ ;
3    $\Phi \leftarrow 0_{1 \times (n+1)}$ ;
4    $\zeta_1^c \leftarrow [h_1, h_2]$ ;
5    $\phi \leftarrow h_3$ ;
6    $\Phi_1 \leftarrow \phi$ ;
7   for  $i \leftarrow 1$  to  $n$  do
8      $\theta \leftarrow \zeta_{i,1}^P$ ;
9      $l \leftarrow \zeta_{i,2}^P$ ;
10     $\phi \leftarrow \text{wrapToPi}(\phi + \theta)$ ;
11     $\zeta_{i+1,1}^c \leftarrow \zeta_{i,1}^c + l \cos(\phi)$ ;
12     $\zeta_{i+1,2}^c \leftarrow \zeta_{i,2}^c + l \sin(\phi)$ ;
13     $\Phi_{i+1} \leftarrow \phi$ ;
14  return  $\zeta^c, \Phi$ ;
```

The Cartesian coordinates of the next node are computed as defined in Eqn.(2)

$$\zeta_{i+1}^c = \zeta_i^c + l_i [\cos(\phi_i) \quad \sin(\phi_i)] \quad (2)$$

This makes the path dependent on the order of the encoded motion commands. Therefore, the resulting trajectory is not an arbitrary set of waypoints, but a sequence of straight movements generated from the relative turning angles.

The path returned by the algorithm S.1 is only the raw decoded path. It may contain more segments than needed because the chromosome can store a maximum number of possible segments. For this reason, it identifies the node used as the terminal cut point and discards the unused nodes. After this, it replaces the last active segment with the corrected segment that connects the selected cut point to the target pose.

2 Statistical Analysis Results

The rest of the document includes tables and figures

Table S.1: LGA experimental results: Kruskal–Wallis test with Dunn–Holm post-hoc procedure.

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	0.0028	0.0072	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	0.0028	0.0028	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	0.0028	0.0002	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	0.0028	0.0133	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	0.0028	0.0099	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	0.0028	0.0000	0.0073	Pop500_Perm1
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	0.0028	0.0000	0.0018	Pop500_Perm2
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	0.0028	0.0000	0.0163	Pop500_Perm3
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	0.0028	0.0000	0.0323	Pop500_Perm4
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	0.0028	0.0000	0.0004	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	0.0028	0.0000	$< 10^{-4}$	Pop500_Perm6
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	0.0072	0.0028	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	0.0072	0.0002	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	0.0072	0.0133	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	0.0072	0.0099	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	0.0072	0.0000	0.0001	Pop500_Perm1
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	0.0072	0.0000	$< 10^{-4}$	Pop500_Perm2
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	0.0072	0.0000	0.0004	Pop500_Perm3
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	0.0072	0.0000	0.0009	Pop500_Perm4
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	0.0072	0.0000	$< 10^{-4}$	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	0.0072	0.0000	$< 10^{-4}$	Pop500_Perm6

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	0.0028	0.0002	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	0.0028	0.0133	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	0.0028	0.0099	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	0.0028	0.0000	0.0010	Pop500_Perm1
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	0.0028	0.0000	0.0002	Pop500_Perm2
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	0.0028	0.0000	0.0022	Pop500_Perm3
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	0.0028	0.0000	0.0049	Pop500_Perm4
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	0.0028	0.0000	$< 10^{-4}$	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	0.0028	0.0000	$< 10^{-4}$	Pop500_Perm6
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	0.0002	0.0133	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	0.0002	0.0099	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	0.0002	0.0000	0.1682	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	0.0002	0.0000	0.0531	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	0.0002	0.0000	0.3111	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	0.0002	0.0000	0.5317	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	0.0002	0.0000	0.0163	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	0.0002	0.0000	0.0011	Pop500_Perm6
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	0.0133	0.0099	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	0.0133	0.0000	0.0003	Pop500_Perm1
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	0.0133	0.0000	$< 10^{-4}$	Pop500_Perm2
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	0.0133	0.0000	0.0006	Pop500_Perm3
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	0.0133	0.0000	0.0014	Pop500_Perm4
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	0.0133	0.0000	$< 10^{-4}$	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	0.0133	0.0000	$< 10^{-4}$	Pop500_Perm6
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	0.0099	0.0000	0.0002	Pop500_Perm1
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	0.0099	0.0000	$< 10^{-4}$	Pop500_Perm2
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	0.0099	0.0000	0.0005	Pop500_Perm3
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	0.0099	0.0000	0.0010	Pop500_Perm4
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	0.0099	0.0000	$< 10^{-4}$	Pop500_Perm5
Task1	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	0.0099	0.0000	$< 10^{-4}$	Pop500_Perm6
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	0.0000	0.0000	1.0000	None
Task1	TerminalError	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	0.0000	0.0000	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	114.8254	114.9093	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	114.8254	114.8479	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	114.8254	114.8676	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	114.8254	114.9394	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	114.8254	114.9256	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	114.8254	114.8494	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	114.8254	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	114.8254	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	114.8254	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	114.8254	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	114.8254	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	114.9093	114.8479	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	114.9093	114.8676	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	114.9093	114.9394	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	114.9093	114.9256	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	114.9093	114.8494	1.0000	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	114.9093	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	114.9093	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	114.9093	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	114.9093	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	114.9093	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	114.8479	114.8676	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	114.8479	114.9394	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	114.8479	114.9256	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	114.8479	114.8494	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	114.8479	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	114.8479	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	114.8479	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	114.8479	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	114.8479	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	114.8676	114.9394	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	114.8676	114.9256	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	114.8676	114.8494	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	114.8676	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	114.8676	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	114.8676	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	114.8676	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	114.8676	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	114.9394	114.9256	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	114.9394	114.8494	0.6638	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	114.9394	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	114.9394	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	114.9394	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	114.9394	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	114.9394	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	114.9256	114.8494	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	114.9256	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	114.9256	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	114.9256	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	114.9256	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	114.9256	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	114.8494	114.2165	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	114.8494	114.8687	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	114.8494	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	114.8494	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	114.8494	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	114.2165	114.8687	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	114.2165	114.8799	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	114.2165	114.8859	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	114.2165	114.8400	$< 10^{-4}$	Pop500_Perm2
Task1	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	114.8687	114.8799	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	114.8687	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	114.8687	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	114.8799	114.8859	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	114.8799	114.8400	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	114.8859	114.8400	1.0000	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop100_Perm2	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop100_Perm3	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop100_Perm4	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop100_Perm5	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop100_Perm6	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm1	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm2	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm3	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm4	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm5	0.0000	0.0000	<i>NaN</i>	None
Task1	Undulation	1.0000	Pop100_Perm1	Pop500_Perm6	0.0000	0.0000	<i>NaN</i>	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

[illegible]

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

[illegible]

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	0.1870	0.0875	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	0.1870	0.2755	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	0.1870	0.0640	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	0.1870	0.1761	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	0.1870	0.0003	0.0095	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	0.1870	0.0007	0.0355	Pop500_Perm2
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	0.1870	0.0006	0.3868	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	0.1870	0.0004	0.0160	Pop500_Perm4
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	0.1870	0.0004	0.0060	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	0.1870	0.0007	0.0168	Pop500_Perm6
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	0.1679	0.0875	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	0.1679	0.2755	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	0.1679	0.0640	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	0.1679	0.1761	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	0.1679	0.0003	0.0003	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	0.1679	0.0007	0.0021	Pop500_Perm2
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	0.1679	0.0006	0.0365	Pop500_Perm3
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	0.1679	0.0004	0.0007	Pop500_Perm4
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	0.1679	0.0004	0.0002	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	0.1679	0.0007	0.0008	Pop500_Perm6
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	0.0875	0.2755	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	0.0875	0.0640	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	0.0875	0.1761	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	0.0875	0.0003	0.0098	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	0.0875	0.0007	0.0365	Pop500_Perm2
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	0.0875	0.0006	0.3946	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	0.0875	0.0004	0.0168	Pop500_Perm4
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	0.0875	0.0004	0.0064	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	0.0875	0.0007	0.0173	Pop500_Perm6
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	0.2755	0.0640	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	0.2755	0.1761	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	0.2755	0.0003	0.0054	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	0.2755	0.0007	0.0214	Pop500_Perm2
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	0.2755	0.0006	0.2563	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	0.2755	0.0004	0.0095	Pop500_Perm4
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	0.2755	0.0004	0.0033	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	0.2755	0.0007	0.0098	Pop500_Perm6
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	0.0640	0.1761	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	0.0640	0.0003	0.0432	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	0.0640	0.0007	0.1571	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	0.0640	0.0006	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	0.0640	0.0004	0.0718	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	0.0640	0.0004	0.0310	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	0.0640	0.0007	0.0746	None
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	0.1761	0.0003	0.0002	Pop500_Perm1
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	0.1761	0.0007	0.0010	Pop500_Perm2
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	0.1761	0.0006	0.0210	Pop500_Perm3
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	0.1761	0.0004	0.0003	Pop500_Perm4
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	0.1761	0.0004	$< 10^{-4}$	Pop500_Perm5
Task2	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	0.1761	0.0007	0.0003	Pop500_Perm6
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	0.0003	0.0007	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	0.0003	0.0006	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	0.0003	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	0.0003	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	0.0003	0.0007	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	0.0007	0.0006	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	0.0007	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	0.0007	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	0.0007	0.0007	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	0.0006	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	0.0006	0.0004	1.0000	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	0.0006	0.0007	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	0.0004	0.0004	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	0.0004	0.0007	1.0000	None
Task2	TerminalError	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	0.0004	0.0007	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	239.9367	239.6585	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	239.9367	241.0564	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	239.9367	242.1870	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	239.9367	243.3172	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	239.9367	241.7584	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	239.9367	239.0196	0.0397	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	239.9367	238.1387	$< 10^{-4}$	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	239.9367	238.9548	0.1800	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	239.9367	239.2318	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	239.9367	238.9306	0.2803	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	239.9367	239.2969	0.9216	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	239.6585	241.0564	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	239.6585	242.1870	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	239.6585	243.3172	0.5674	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	239.6585	241.7584	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	239.6585	239.0196	0.2803	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	239.6585	238.1387	0.0006	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	239.6585	238.9548	0.9042	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	239.6585	239.2318	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	239.6585	238.9306	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	239.6585	239.2969	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	241.0564	242.1870	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	241.0564	243.3172	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	241.0564	241.7584	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	241.0564	239.0196	0.0209	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	241.0564	238.1387	$< 10^{-4}$	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	241.0564	238.9548	0.1068	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	241.0564	239.2318	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	241.0564	238.9306	0.1760	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	241.0564	239.2969	0.6539	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	242.1870	243.3172	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	242.1870	241.7584	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	242.1870	239.0196	0.0006	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	242.1870	238.1387	$< 10^{-4}$	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	242.1870	238.9548	0.0051	Pop500_Perm3
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	242.1870	239.2318	0.6346	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	242.1870	238.9306	0.0095	Pop500_Perm5
Task2	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	242.1870	239.2969	0.0576	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	243.3172	241.7584	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	243.3172	239.0196	$< 10^{-4}$	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	243.3172	238.1387	$< 10^{-4}$	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	243.3172	238.9548	0.0002	Pop500_Perm3
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	243.3172	239.2318	0.0822	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	243.3172	238.9306	0.0004	Pop500_Perm5
Task2	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	243.3172	239.2969	0.0040	Pop500_Perm6
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	241.7584	239.0196	0.0010	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	241.7584	238.1387	$< 10^{-4}$	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	241.7584	238.9548	0.0079	Pop500_Perm3
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	241.7584	239.2318	0.7804	None
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	241.7584	238.9306	0.0140	Pop500_Perm5
Task2	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	241.7584	239.2969	0.0822	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	239.0196	238.1387	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	239.0196	238.9548	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	239.0196	239.2318	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	239.0196	238.9306	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	239.0196	239.2969	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	238.1387	238.9548	0.9216	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	238.1387	239.2318	0.0108	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	238.1387	238.9306	0.6613	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	238.1387	239.2969	0.1800	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	238.9548	239.2318	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	238.9548	238.9306	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	238.9548	239.2969	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	239.2318	238.9306	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	239.2318	239.2969	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	238.9306	239.2969	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	1.0000	2.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	1.0000	2.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	1.0000	1.0000	0.7961	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	1.0000	1.0000	0.4976	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	1.0000	1.0000	0.2315	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	1.0000	1.0000	0.5773	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	1.0000	1.0000	0.1839	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	1.0000	1.0000	0.0107	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	1.0000	1.0000	0.0107	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	2.0000	2.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	2.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	2.0000	1.0000	0.0982	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	2.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	2.0000	1.0000	0.0524	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	2.0000	1.0000	0.0189	Pop500_Perm2
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	2.0000	1.0000	0.0682	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	2.0000	1.0000	0.0141	Pop500_Perm4
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	2.0000	1.0000	0.0004	Pop500_Perm5
Task2	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	2.0000	1.0000	0.0004	Pop500_Perm6
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	2.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	2.0000	1.0000	0.1230	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	2.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	2.0000	1.0000	0.0682	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	2.0000	1.0000	0.0248	Pop500_Perm2
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	2.0000	1.0000	0.0866	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	2.0000	1.0000	0.0189	Pop500_Perm4
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	2.0000	1.0000	0.0005	Pop500_Perm5
Task2	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	2.0000	1.0000	0.0005	Pop500_Perm6
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	1.0000	1.0000	0.4976	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	1.0000	1.0000	0.4976	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	1.0000	1.0000	0.4700	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	1.0000	1.0000	0.2654	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	1.0000	1.0000	0.1152	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	1.0000	1.0000	0.3304	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	1.0000	1.0000	0.0916	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	1.0000	1.0000	0.0042	None
Task2	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	1.0000	1.0000	0.0042	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	1.0000	1.0000	1.0000	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	6.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	6.0000	5.0000	0.0895	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	6.0000	5.0000	0.0148	Pop100_Perm4
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	6.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	6.0000	6.0000	0.9968	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm1
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm2
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm3
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm5
Task2	Segments	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	6.0000	5.0000	$< 10^{-4}$	Pop500_Perm6
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	6.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	6.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	6.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	6.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	6.0000	5.0000	0.1279	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	6.0000	5.0000	0.0086	Pop500_Perm2
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	6.0000	5.0000	0.0086	Pop500_Perm3
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	6.0000	5.0000	0.0010	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	6.0000	5.0000	0.1279	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	6.0000	5.0000	0.0003	Pop500_Perm6
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	5.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	5.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	5.0000	5.0000	0.4333	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	5.0000	5.0000	0.4333	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	5.0000	5.0000	0.1023	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	5.0000	5.0000	0.0426	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	5.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	5.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	5.0000	5.0000	0.4333	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	5.0000	5.0000	0.2235	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	6.0000	6.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	6.0000	5.0000	0.2235	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	6.0000	5.0000	0.0172	Pop500_Perm2
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	6.0000	5.0000	0.0172	Pop500_Perm3
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	6.0000	5.0000	0.0022	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	6.0000	5.0000	0.2235	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	6.0000	5.0000	0.0007	Pop500_Perm6
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	6.0000	5.0000	0.3226	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	6.0000	5.0000	0.0286	Pop500_Perm2

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	6.0000	5.0000	0.0286	Pop500_Perm3
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	6.0000	5.0000	0.0039	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	6.0000	5.0000	0.3226	None
Task2	Segments	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	6.0000	5.0000	0.0013	Pop500_Perm6
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	5.0000	5.0000	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	0.2689	0.2419	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	0.2689	0.1970	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	0.2689	0.3463	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	0.2689	0.2630	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	0.2689	0.2373	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	0.2689	0.0012	$< 10^{-4}$	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	0.2689	0.0017	0.0083	Pop500_Perm2
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	0.2689	0.0026	0.0080	Pop500_Perm3
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	0.2689	0.0079	0.0663	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	0.2689	0.1364	0.8281	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	0.2689	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	0.2419	0.1970	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	0.2419	0.3463	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	0.2419	0.2630	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	0.2419	0.2373	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	0.2419	0.0012	0.0091	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	0.2419	0.0017	0.2543	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	0.2419	0.0026	0.2511	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	0.2419	0.0079	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	0.2419	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	0.2419	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	0.1970	0.3463	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	0.1970	0.2630	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	0.1970	0.2373	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	0.1970	0.0012	0.0006	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	0.1970	0.0017	0.0335	Pop500_Perm2
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	0.1970	0.0026	0.0328	Pop500_Perm3
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	0.1970	0.0079	0.2298	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	0.1970	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	0.1970	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	0.3463	0.2630	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	0.3463	0.2373	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	0.3463	0.0012	$< 10^{-4}$	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	0.3463	0.0017	0.0017	Pop500_Perm2
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	0.3463	0.0026	0.0016	Pop500_Perm3
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	0.3463	0.0079	0.0182	Pop500_Perm4
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	0.3463	0.1364	0.2915	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	0.3463	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	0.2630	0.2373	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	0.2630	0.0012	0.0001	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	0.2630	0.0017	0.0096	Pop500_Perm2
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	0.2630	0.0026	0.0093	Pop500_Perm3

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	0.2630	0.0079	0.0768	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	0.2630	0.1364	0.9183	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	0.2630	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	0.2373	0.0012	0.0008	Pop500_Perm1
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	0.2373	0.0017	0.0413	Pop500_Perm2
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	0.2373	0.0026	0.0405	Pop500_Perm3
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	0.2373	0.0079	0.2651	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	0.2373	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	0.2373	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	0.0012	0.0017	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	0.0012	0.0026	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	0.0012	0.0079	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	0.0012	0.1364	0.5010	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	0.0012	0.0847	0.0536	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	0.0017	0.0026	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	0.0017	0.0079	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	0.0017	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	0.0017	0.0847	0.9183	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	0.0026	0.0079	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	0.0026	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	0.0026	0.0847	0.9183	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	0.0079	0.1364	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	0.0079	0.0847	1.0000	None
Task3	TerminalError	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	0.1364	0.0847	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	208.7650	208.8447	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	208.7650	209.6809	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	208.7650	209.5999	0.2958	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	208.7650	210.2616	0.0040	Pop100_Perm1
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	208.7650	209.9329	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	208.7650	209.0405	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	208.7650	207.6198	0.0285	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	208.7650	208.4441	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	208.7650	208.6828	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	208.7650	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	208.7650	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	208.8447	209.6809	0.0760	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	208.8447	209.5999	0.0063	Pop100_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	208.8447	210.2616	$< 10^{-4}$	Pop100_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	208.8447	209.9329	0.0760	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	208.8447	209.0405	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	208.8447	207.6198	0.8348	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	208.8447	208.4441	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	208.8447	208.6828	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	208.8447	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	208.8447	209.5132	0.0726	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	209.6809	209.5999	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	209.6809	210.2616	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	209.6809	209.9329	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	209.6809	209.0405	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	209.6809	207.6198	$< 10^{-4}$	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	209.6809	208.4441	0.0094	Pop500_Perm3
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	209.6809	208.6828	0.6513	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	209.6809	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	209.6809	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	209.5999	210.2616	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	209.5999	209.9329	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	209.5999	209.0405	0.2958	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	209.5999	207.6198	$< 10^{-4}$	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	209.5999	208.4441	0.0005	Pop500_Perm3
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	209.5999	208.6828	0.0854	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	209.5999	209.1657	0.8513	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task3	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	209.5999	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	210.2616	209.9329	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	210.2616	209.0405	0.0040	Pop500_Perm1
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	210.2616	207.6198	$< 10^{-4}$	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	210.2616	208.4441	$< 10^{-4}$	Pop500_Perm3
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	210.2616	208.6828	0.0007	Pop500_Perm4
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	210.2616	209.1657	0.0198	Pop500_Perm5
Task3	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	210.2616	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	209.9329	209.0405	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	209.9329	207.6198	$< 10^{-4}$	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	209.9329	208.4441	0.0094	Pop500_Perm3
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	209.9329	208.6828	0.6513	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	209.9329	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	209.9329	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	209.0405	207.6198	0.0285	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	209.0405	208.4441	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	209.0405	208.6828	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	209.0405	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	209.0405	209.5132	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	207.6198	208.4441	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	207.6198	208.6828	0.1038	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	207.6198	209.1657	0.0060	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	207.6198	209.5132	$< 10^{-4}$	Pop500_Perm2
Task3	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	208.4441	208.6828	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	208.4441	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	208.4441	209.5132	0.0089	Pop500_Perm3
Task3	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	208.6828	209.1657	1.0000	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	208.6828	209.5132	0.6332	None
Task3	PathLength	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	209.1657	209.5132	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	3.0000	3.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	3.0000	3.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm1	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm2	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm3	3.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm4	3.0000	1.0000	0.3689	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm5	3.0000	1.0000	0.0323	Pop500_Perm5
Task3	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop500_Perm6	3.0000	1.0000	0.0032	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	3.0000	3.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm1	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm2	3.0000	2.0000	0.0852	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm3	3.0000	1.0000	0.0537	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm4	3.0000	1.0000	0.0039	Pop500_Perm4
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm5	3.0000	1.0000	0.0001	Pop500_Perm5
Task3	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop500_Perm6	3.0000	1.0000	$< 10^{-4}$	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm1	3.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm2	3.0000	2.0000	0.7344	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm3	3.0000	1.0000	0.5069	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm4	3.0000	1.0000	0.0614	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm5	3.0000	1.0000	0.0035	Pop500_Perm5
Task3	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop500_Perm6	3.0000	1.0000	0.0002	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	2.0000	2.0000	1.0000	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm1	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm2	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm3	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm4	2.0000	1.0000	0.1865	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm5	2.0000	1.0000	0.0136	Pop500_Perm5
Task3	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop500_Perm6	2.0000	1.0000	0.0012	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm1	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm2	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm3	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm4	2.0000	1.0000	0.8608	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm5	2.0000	1.0000	0.0953	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop500_Perm6	2.0000	1.0000	0.0118	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm1	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm2	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm3	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm4	2.0000	1.0000	0.1865	None
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm5	2.0000	1.0000	0.0136	Pop500_Perm5
Task3	Undulation	$< 10^{-4}$	Pop100_Perm6	Pop500_Perm6	2.0000	1.0000	0.0012	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	2.0000	2.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	2.0000	1.0000	0.5652	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	2.0000	1.0000	0.0560	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	2.0000	1.0000	0.0061	Pop500_Perm6
Task3	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	2.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop100_Perm2	8.0000	9.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop100_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop100_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop100_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop100_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm1	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm4	8.0000	8.0000	0.4463	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm1	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm2	Pop100_Perm3	9.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm2	Pop100_Perm4	9.0000	8.0000	0.0491	Pop100_Perm4
Task3	Segments	0.0006	Pop100_Perm2	Pop100_Perm5	9.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm2	Pop100_Perm6	9.0000	8.0000	0.7181	None
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm1	9.0000	8.0000	0.2325	None
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm2	9.0000	8.0000	0.1659	None
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm3	9.0000	8.0000	0.0113	Pop500_Perm3
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm4	9.0000	8.0000	0.0031	Pop500_Perm4
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm5	9.0000	8.0000	0.8289	None
Task3	Segments	0.0006	Pop100_Perm2	Pop500_Perm6	9.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop100_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop100_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop100_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm1	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm3	8.0000	8.0000	0.4180	None

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Table S.1: Kruskal-Wallis and Pairwise Dunn-Holm Comparisons (Continued)

Task	Metric	KW p	Config i	Config j	Median i	Median j	Dunn p_{adj}	Winner
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm4	8.0000	8.0000	0.1659	None
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm3	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop100_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop100_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm1	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm4	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm5	Pop100_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm1	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm3	8.0000	8.0000	0.4552	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm4	8.0000	8.0000	0.1837	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm5	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm1	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop100_Perm6	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm1	Pop500_Perm2	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm1	Pop500_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm1	Pop500_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm1	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm1	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm2	Pop500_Perm3	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm2	Pop500_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm2	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm2	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm3	Pop500_Perm4	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm3	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm3	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm4	Pop500_Perm5	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm4	Pop500_Perm6	8.0000	8.0000	1.0000	None
Task3	Segments	0.0006	Pop500_Perm5	Pop500_Perm6	8.0000	8.0000	1.0000	None

Table S.2: NSGA-III-RRT* experimental results: Friedman test and Wilcoxon post-hoc pairwise comparisons

Task	Metric	Friedman p	Config 1	Config 2	Median 1	Median 2	Wilcoxon p_{adj}	Winner
Task1	TerminalError	0.0477	Pop100_Perm1	Pop100_Perm2	0.1512	0.1725	0.5127	None
Task1	TerminalError	0.0477	Pop100_Perm1	Pop100_Perm3	0.1512	0.2966	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm1	Pop100_Perm4	0.1512	0.2898	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm1	Pop100_Perm5	0.1512	0.1512	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm1	Pop100_Perm6	0.1512	0.2506	0.6321	None
Task1	TerminalError	0.0477	Pop100_Perm2	Pop100_Perm3	0.1725	0.2966	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm2	Pop100_Perm4	0.1725	0.2898	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm2	Pop100_Perm5	0.1725	0.1512	0.5127	None
Task1	TerminalError	0.0477	Pop100_Perm2	Pop100_Perm6	0.1725	0.2506	0.7056	None
Task1	TerminalError	0.0477	Pop100_Perm3	Pop100_Perm4	0.2966	0.2898	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm3	Pop100_Perm5	0.2966	0.1512	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm3	Pop100_Perm6	0.2966	0.2506	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm4	Pop100_Perm5	0.2898	0.1512	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm4	Pop100_Perm6	0.2898	0.2506	1.0000	None
Task1	TerminalError	0.0477	Pop100_Perm5	Pop100_Perm6	0.1512	0.2506	0.6321	None

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Table S.2: Friedman Global Tests and Pairwise Wilcoxon Post-Hoc Comparisons (Continued)

Task	Metric	Friedman p	Config 1	Config 2	Median 1	Median 2	Wilcoxon p_{adj}	Winner
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	113.4016	113.4668	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	113.4016	115.8228	0.0016	Pop100_Perm1
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	113.4016	115.8516	0.0016	Pop100_Perm1
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	113.4016	113.4016	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	113.4016	114.9367	0.0229	Pop100_Perm1
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	113.4668	115.8228	0.0016	Pop100_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	113.4668	115.8516	0.0016	Pop100_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	113.4668	113.4016	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	113.4668	114.9367	0.0229	Pop100_Perm2
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	115.8228	115.8516	1.0000	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	115.8228	113.4016	0.0016	Pop100_Perm5
Task1	PathLength	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	115.8228	114.9367	0.0977	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	115.8516	113.4016	0.0016	Pop100_Perm5
Task1	PathLength	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	115.8516	114.9367	0.0938	None
Task1	PathLength	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	113.4016	114.9367	0.0229	Pop100_Perm5
Task1	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	0.0000	0.0000	0.1172	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	0.0000	0.0000	0.1875	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	0.0000	0.0000	0.1875	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	0.0000	0.0000	0.5625	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	0.0000	0.0000	1.0000	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	0.0000	0.0000	0.1172	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	0.0000	0.0000	0.1172	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	0.0000	0.0000	0.1875	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	0.0000	0.0000	0.1875	None
Task1	Undulation	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	0.0000	0.0000	1.0000	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm2	4.0000	4.0000	1.0000	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm3	4.0000	3.0000	0.0002	Pop100_Perm3
Task1	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm4	4.0000	3.0000	0.0002	Pop100_Perm4
Task1	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm5	4.0000	4.0000	1.0000	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm1	Pop100_Perm6	4.0000	3.0000	0.0044	Pop100_Perm6
Task1	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm3	4.0000	3.0000	0.0002	Pop100_Perm3
Task1	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm4	4.0000	3.0000	0.0002	Pop100_Perm4
Task1	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm5	4.0000	4.0000	1.0000	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm2	Pop100_Perm6	4.0000	3.0000	0.0068	Pop100_Perm6
Task1	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm4	3.0000	3.0000	1.0000	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm5	3.0000	4.0000	0.0002	Pop100_Perm3
Task1	Segments	$< 10^{-4}$	Pop100_Perm3	Pop100_Perm6	3.0000	3.0000	0.0938	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm5	3.0000	4.0000	0.0002	Pop100_Perm4
Task1	Segments	$< 10^{-4}$	Pop100_Perm4	Pop100_Perm6	3.0000	3.0000	0.0938	None
Task1	Segments	$< 10^{-4}$	Pop100_Perm5	Pop100_Perm6	4.0000	3.0000	0.0044	Pop100_Perm6
Task2	PathLength	0.0007	Pop100_Perm1	Pop100_Perm2	233.4583	233.4583	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm1	Pop100_Perm3	233.4583	237.5256	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm1	Pop100_Perm4	233.4583	239.6183	0.3397	None
Task2	PathLength	0.0007	Pop100_Perm1	Pop100_Perm5	233.4583	239.6183	0.0446	Pop100_Perm1
Task2	PathLength	0.0007	Pop100_Perm1	Pop100_Perm6	233.4583	240.8528	0.0446	Pop100_Perm1
Task2	PathLength	0.0007	Pop100_Perm2	Pop100_Perm3	233.4583	237.5256	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm2	Pop100_Perm4	233.4583	239.6183	0.3625	None
Task2	PathLength	0.0007	Pop100_Perm2	Pop100_Perm5	233.4583	239.6183	0.0446	Pop100_Perm2
Task2	PathLength	0.0007	Pop100_Perm2	Pop100_Perm6	233.4583	240.8528	0.0446	Pop100_Perm2
Task2	PathLength	0.0007	Pop100_Perm3	Pop100_Perm4	237.5256	239.6183	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm3	Pop100_Perm5	237.5256	239.6183	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm3	Pop100_Perm6	237.5256	240.8528	0.2445	None
Task2	PathLength	0.0007	Pop100_Perm4	Pop100_Perm5	239.6183	239.6183	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm4	Pop100_Perm6	239.6183	240.8528	1.0000	None
Task2	PathLength	0.0007	Pop100_Perm5	Pop100_Perm6	239.6183	240.8528	1.0000	None
Task2	Undulation	0.0009	Pop100_Perm1	Pop100_Perm2	3.0000	3.0000	1.0000	None

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Table S.2: Friedman Global Tests and Pairwise Wilcoxon Post-Hoc Comparisons (Continued)

Task	Metric	Friedman p	Config 1	Config 2	Median 1	Median 2	Wilcoxon p_{adj}	Winner
Task2	Undulation	0.0009	Pop100_Perm1	Pop100_Perm3	3.0000	2.0000	0.6982	None
Task2	Undulation	0.0009	Pop100_Perm1	Pop100_Perm4	3.0000	2.0000	0.3395	None
Task2	Undulation	0.0009	Pop100_Perm1	Pop100_Perm5	3.0000	1.0000	0.0700	None
Task2	Undulation	0.0009	Pop100_Perm1	Pop100_Perm6	3.0000	1.0000	0.0700	None
Task2	Undulation	0.0009	Pop100_Perm2	Pop100_Perm3	3.0000	2.0000	0.6802	None
Task2	Undulation	0.0009	Pop100_Perm2	Pop100_Perm4	3.0000	2.0000	0.2104	None
Task2	Undulation	0.0009	Pop100_Perm2	Pop100_Perm5	3.0000	1.0000	0.0663	None
Task2	Undulation	0.0009	Pop100_Perm2	Pop100_Perm6	3.0000	1.0000	0.0663	None
Task2	Undulation	0.0009	Pop100_Perm3	Pop100_Perm4	2.0000	2.0000	0.6802	None
Task2	Undulation	0.0009	Pop100_Perm3	Pop100_Perm5	2.0000	1.0000	0.1518	None
Task2	Undulation	0.0009	Pop100_Perm3	Pop100_Perm6	2.0000	1.0000	0.1518	None
Task2	Undulation	0.0009	Pop100_Perm4	Pop100_Perm5	2.0000	1.0000	0.6982	None
Task2	Undulation	0.0009	Pop100_Perm4	Pop100_Perm6	2.0000	1.0000	0.6982	None
Task2	Undulation	0.0009	Pop100_Perm5	Pop100_Perm6	1.0000	1.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm1	Pop100_Perm2	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm1	Pop100_Perm3	5.0000	5.0000	0.8350	None
Task2	Segments	0.0025	Pop100_Perm1	Pop100_Perm4	5.0000	5.0000	0.8350	None
Task2	Segments	0.0025	Pop100_Perm1	Pop100_Perm5	5.0000	6.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm1	Pop100_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm2	Pop100_Perm3	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm2	Pop100_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm2	Pop100_Perm5	5.0000	6.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm2	Pop100_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm3	Pop100_Perm4	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm3	Pop100_Perm5	5.0000	6.0000	0.9902	None
Task2	Segments	0.0025	Pop100_Perm3	Pop100_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm4	Pop100_Perm5	5.0000	6.0000	0.9902	None
Task2	Segments	0.0025	Pop100_Perm4	Pop100_Perm6	5.0000	5.0000	1.0000	None
Task2	Segments	0.0025	Pop100_Perm5	Pop100_Perm6	6.0000	5.0000	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	238.1723	238.1723	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	238.1723	251.0914	0.0063	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	238.1723	251.0914	0.0063	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	238.1723	246.4502	0.0241	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	238.1723	252.2412	0.0063	Pop500_Perm1
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	238.1723	251.0914	0.0063	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	238.1723	251.0914	0.0063	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	238.1723	246.4502	0.0241	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	238.1723	252.2412	0.0063	Pop500_Perm2
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	251.0914	251.0914	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	251.0914	246.4502	0.3777	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	251.0914	252.2412	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	251.0914	246.4502	0.3777	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	251.0914	252.2412	1.0000	None
Task2	PathLength	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	246.4502	252.2412	0.0348	Pop500_Perm5
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	2.0000	2.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	2.0000	1.0000	0.2582	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	2.0000	1.0000	0.2471	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	2.0000	1.0000	0.0732	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	2.0000	1.0000	0.0732	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	2.0000	1.0000	0.2582	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	2.0000	1.0000	0.2471	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	2.0000	1.0000	0.0732	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	2.0000	1.0000	0.0732	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Undulation	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm2	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm3	5.0000	4.0000	0.0160	Pop500_Perm3

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Table S.2: Friedman Global Tests and Pairwise Wilcoxon Post-Hoc Comparisons (Continued)

Task	Metric	Friedman p	Config 1	Config 2	Median 1	Median 2	Wilcoxon p_{adj}	Winner
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm4	5.0000	4.0000	0.0160	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm1	Pop500_Perm6	5.0000	4.0000	0.0859	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm3	5.0000	4.0000	0.0160	Pop500_Perm3
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm4	5.0000	4.0000	0.0160	Pop500_Perm4
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm5	5.0000	5.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm2	Pop500_Perm6	5.0000	4.0000	0.0859	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm4	4.0000	4.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm5	4.0000	5.0000	0.0859	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm3	Pop500_Perm6	4.0000	4.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm5	4.0000	5.0000	0.0859	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm4	Pop500_Perm6	4.0000	4.0000	1.0000	None
Task2	Segments	$< 10^{-4}$	Pop500_Perm5	Pop500_Perm6	5.0000	4.0000	0.1230	None
Task3	TerminalError	0.0002	Pop100_Perm1	Pop100_Perm2	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm1	Pop100_Perm3	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm1	Pop100_Perm4	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm1	Pop100_Perm5	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm1	Pop100_Perm6	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm2	Pop100_Perm3	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm2	Pop100_Perm4	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm2	Pop100_Perm5	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm2	Pop100_Perm6	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm3	Pop100_Perm4	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm3	Pop100_Perm5	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm3	Pop100_Perm6	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm4	Pop100_Perm5	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm4	Pop100_Perm6	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0002	Pop100_Perm5	Pop100_Perm6	0.3719	0.3719	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm1	Pop500_Perm2	0.3731	0.3731	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm1	Pop500_Perm3	0.3731	0.3834	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm1	Pop500_Perm4	0.3731	0.3818	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm1	Pop500_Perm5	0.3731	0.3270	0.9375	None
Task3	TerminalError	0.0062	Pop500_Perm1	Pop500_Perm6	0.3731	0.3270	0.9375	None
Task3	TerminalError	0.0062	Pop500_Perm2	Pop500_Perm3	0.3731	0.3834	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm2	Pop500_Perm4	0.3731	0.3818	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm2	Pop500_Perm5	0.3731	0.3270	0.9375	None
Task3	TerminalError	0.0062	Pop500_Perm2	Pop500_Perm6	0.3731	0.3270	0.9375	None
Task3	TerminalError	0.0062	Pop500_Perm3	Pop500_Perm4	0.3834	0.3818	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm3	Pop500_Perm5	0.3834	0.3270	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm3	Pop500_Perm6	0.3834	0.3270	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm4	Pop500_Perm5	0.3818	0.3270	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm4	Pop500_Perm6	0.3818	0.3270	1.0000	None
Task3	TerminalError	0.0062	Pop500_Perm5	Pop500_Perm6	0.3270	0.3270	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm1	Pop500_Perm2	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm1	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm1	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm1	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm1	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm2	Pop500_Perm3	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm2	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm2	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm2	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm3	Pop500_Perm4	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm3	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm3	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm4	Pop500_Perm5	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm4	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Undulation	0.0344	Pop500_Perm5	Pop500_Perm6	1.0000	1.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm1	Pop500_Perm2	7.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm1	Pop500_Perm3	7.0000	6.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm1	Pop500_Perm4	7.0000	6.0000	1.0000	None

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Table S.2: Friedman Global Tests and Pairwise Wilcoxon Post-Hoc Comparisons (Continued)

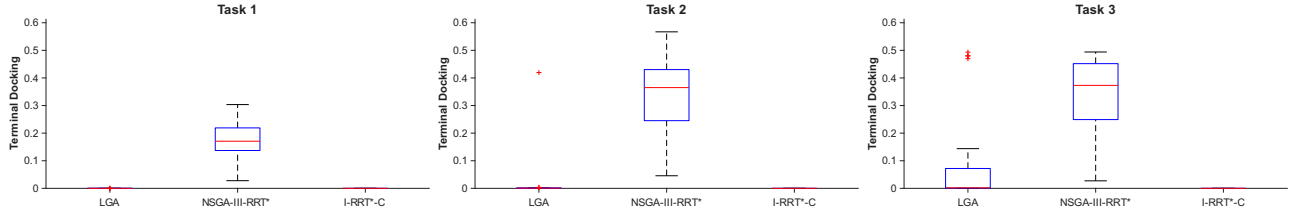
Task	Metric	Friedman p	Config 1	Config 2	Median 1	Median 2	Wilcoxon p_{adj}	Winner
Task3	Segments	0.0104	Pop500_Perm1	Pop500_Perm5	7.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm1	Pop500_Perm6	7.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm2	Pop500_Perm3	7.0000	6.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm2	Pop500_Perm4	7.0000	6.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm2	Pop500_Perm5	7.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm2	Pop500_Perm6	7.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm3	Pop500_Perm4	6.0000	6.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm3	Pop500_Perm5	6.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm3	Pop500_Perm6	6.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm4	Pop500_Perm5	6.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm4	Pop500_Perm6	6.0000	7.0000	1.0000	None
Task3	Segments	0.0104	Pop500_Perm5	Pop500_Perm6	7.0000	7.0000	1.0000	None

Table S.3: LGA vs. NSGA-III vs I-RRT*-C: Comparison of Best-Case Execution Runs

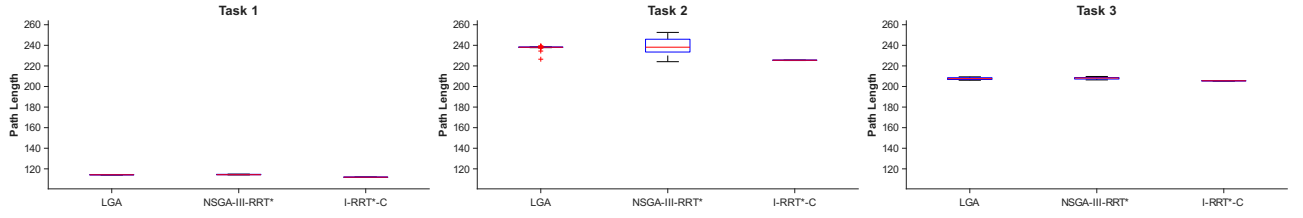
Task	Algorithm	Population	Permutation (Run)	Terminal Docking	Path Length	Undulation	Segment
Task1	LGA	Pop100	Perm4 (12)	0.00	114.6	0	3
	LGA	Pop500	Perm3 (24)	0.00	114.7	0	3
	NSGAIII	Pop100	Perm1 (6)	0.01	114.2	0	4
	NSGAIII	Pop500	Perm1 (1)	0.02	114.7	0	3
	RRT	x	x (x)	0.00	111.8	0	100
Task2	LGA	Pop100	Perm3 (4)	0.00	239.8	2	5
	LGA	Pop500	Perm6 (2)	0.00	238.8	1	5
	NSGAIII	Pop100	Perm5 (2)	0.04	239.2	1	5
	NSGAIII	Pop500	Perm4 (7)	0.04	252.4	1	4
	RRT	x	x (x)	0.00	225.2	17	100
Task3	LGA	Pop100	Perm6 (23)	0.00	207.2	1	8
	LGA	Pop500	Perm5 (9)	0.00	207.9	1	8
	NSGAIII	Pop100	Perm1 (3)	0.00	209.3	1	7
	NSGAIII	Pop500	Perm1 (12)	0.02	207.6	1	7
	RRT	x	x (x)	0.00	205.2	8	100

Table S.4: LGA vs. NSGA-III vs I-RRT*-C: Comparison of Worst-Case Execution Runs

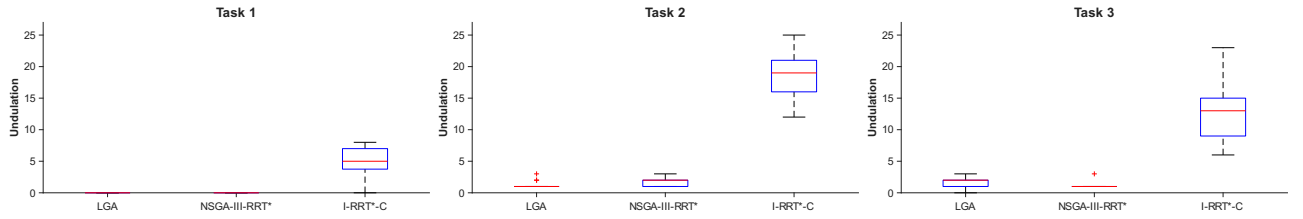
Task	Algorithm	Population	Permutation (Run)	Terminal Docking	Path Length	Undulation	Segment
Task1	LGA	Pop100	Perm1 (13)	0.3992	114.2	0	3
	LGA	Pop500	Perm1 (11)	0.0002	114.7	0	3
	NSGAIII	Pop100	Perm3 (8)	0.4767	115.3	0	3
	NSGAIII	Pop500	Perm1 (23)	0.3035	114.2	0	3
	RRT	x	x (x)	0.00	111.9	7	100
Task2	LGA	Pop100	Perm3 (15)	0.9852	232.6	3	5
	LGA	Pop500	Perm3 (19)	0.4937	239.2	2	5
	NSGAIII	Pop100	Perm1 (9)	1.1412	251.1	2	6
	NSGAIII	Pop500	Perm1 (18)	0.5670	234.7	3	6
	RRT	x	x (x)	0.00	225.6	22	100
Task3	LGA	Pop100	Perm4 (21)	2.4143	209.7	3	8
	LGA	Pop500	Perm6 (15)	0.4978	209.2	1	8
	NSGAIII	Pop100	Perm1 (25)	0.7452	216.7	3	7
	NSGAIII	Pop500	Perm1 (24)	0.4940	206.4	1	7
	RRT	x	x (x)	0.00	205.7	13	100



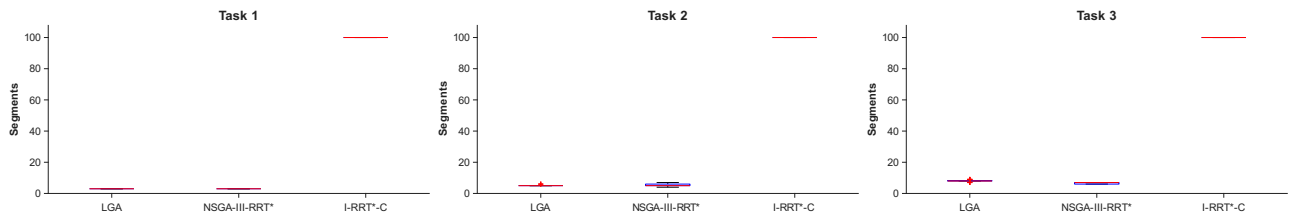
(a) Terminal Docking Error f_1



(b) Path Length f_2



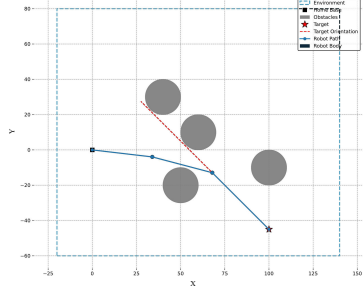
(c) Undulation f_3



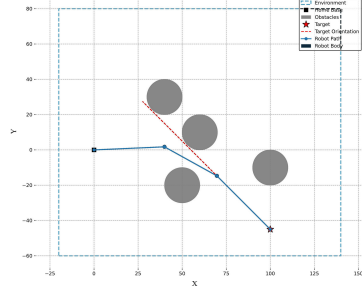
(d) Segments f_4

Figure S.1: Results of the four objective-function values for each algorithm across the three tasks.

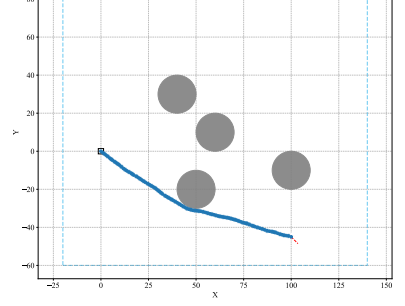
F1 (TD): 0.00, F2 (Len): 114.7, F3 (Und): 0.00, F4 (# Seg): 3



F1 (TD): 0.00, F2 (Len): 116.8, F3 (Und): 1.00, F4 (# Seg): 3

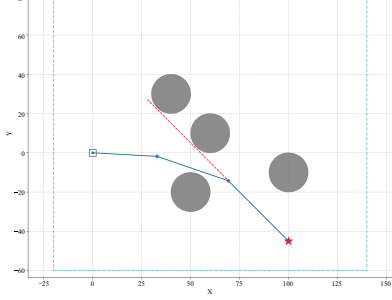


F1 (TD): 0.00, F2 (Len): 111.8, F3 (Und): 0.00, F4 (# Seg): 100

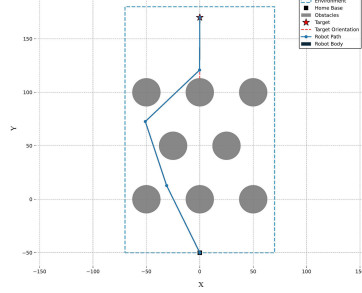


(a) Best LGA, NSGA-III-RRT* and I-RRT*-C for Task 1

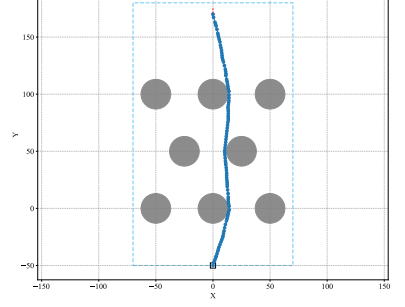
F1 (TD): 0.00, F2 (Len): 114.9, F3 (Und): 0.00, F4 (# Seg): 3



F1 (TD): 0.04, F2 (Len): 252.4, F3 (Und): 1.00, F4 (# Seg): 4

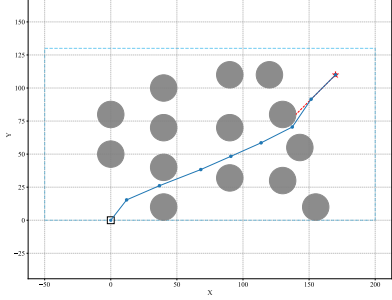


F1 (TD): 0.00, F2 (Len): 225.2, F3 (Und): 17.00, F4 (# Seg): 100

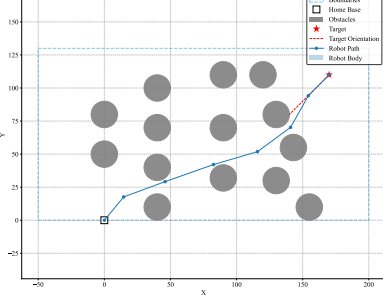


(b) Best LGA, NSGA-III-RRT* and I-RRT*-C for Task 2

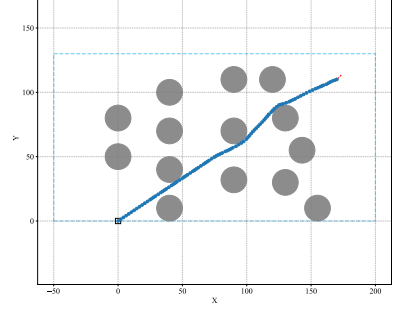
F1 (TD): 0.00, F2 (Len): 208.0, F3 (Und): 1.00, F4 (# Seg): 8



F1 (TD): 0.02, F2 (Len): 210.6, F3 (Und): 1.00, F4 (# Seg): 7



F1 (TD): 0.00, F2 (Len): 205.2, F3 (Und): 8.00, F4 (# Seg): 100



(c) Best LGA, NSGA-III-RRT* and I-RRT*-C results for Task 3

Figure S.2: Best LGA, NSGA-III-RRT* and I-RRT*-C results for Tasks 1–3, respectively

References

- [1] F. Stroppa, Design optimizer for planar soft-growing robot manipulators, Engineering Applications of Artificial Intelligence 130 (2024) 107693.